

Akshay Deshpande

DevOps Engineer

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SUMMARY

DevOps Engineer with 3+ years of experience building and operating cloud-native infrastructure on AWS and Azure. Experienced in CI/CD, Kubernetes/EKS, Helm, Docker Swarm, Terraform, observability, and DevSecOps. Global Siemens AI Hackathon winner who mentors engineers and drives reliable, secure delivery.

EXPERIENCE

Siemens — DevOps Engineer

Nov 2025 – Present

Goa, India

- Build and maintain Azure Pipelines in YAML for CI/CD across multiple applications, automating build, test, and deployment stages and reducing manual release effort.
- Author Helm charts and operate AWS EKS clusters, managing workload scaling, resource requests and limits, pod lifecycle, and troubleshooting.
- Led end-to-end deployment of Timesheet and AI Suite on an on-prem Ubuntu setup using WSL2, from architecture through production rollout.
- Implemented Docker Swarm and Portainer for resilient multi-container deployments; managed Azure Repos and a self-hosted PostgreSQL database with automated continuous backups.
- Deployed Grafana, Loki, and Prometheus for centralized metrics, logs, and alerting across on-prem services.
- Won the global Siemens AI Hackathon with an AI-driven solution; mentor two interns in DevOps, containers, CI/CD, and cloud fundamentals.

Zaggle — Associate DevOps Engineer

Apr 2023 – Nov 2025

Hyderabad, India

- Led a Terraform-driven ECS migration across AWS regions for disaster recovery, achieving a zero-downtime transition and automated re-provisioning.
- Built a production-grade EKS proof of concept with ECR, Auto Scaling Groups, VPC, ALB, GitHub Actions CI, and ArgoCD GitOps delivery.
- Automated deployment pipelines for 60+ microservices across four business verticals, increasing release velocity by 40%.
- Developed reusable Terraform modules for multi-region AWS infrastructure, including Hyderabad DR, accelerating provisioning by 40%.
- Integrated SonarQube into CI, reducing critical vulnerabilities by 25%; embedded security scanning, linting, testing, and approval workflows in Jenkins and GitHub Actions.
- Enforced least-privilege IAM and DevSecOps baselines; implemented fault-tolerant autoscaling patterns that contributed to 99.99% availability.

SELECTED PROJECTS

eksp — Production-Grade EKS Platform

- Engineered a Terraform-provisioned EKS platform targeting 1M requests/minute sustained (~17k RPS) and 3× spike absorption, expressed as executable k6 smoke, ramp, spike, and soak thresholds mapped to SLOs.
- Implemented Karpenter with spot-first NodePools, on-demand fallback, consolidation budgets, and a Graviton system pool; added RPS-based HPA via prometheus-adapter.
- Enforced strict mTLS and authorization policies with Istio ambient mesh. Delivered GitOps through ArgoCD app-of-apps, sync waves, and environment promotion.
- Built observability with kube-prometheus-stack, RED dashboards, and multi-window burn-rate alerts; shipped *costwatch*, a Go and React FinOps app tracing hourly AWS spend.
- Hardened the platform with OIDC-only CI, Pod Identity, restricted Pod Security Standards, default-deny networking, KMS, IMDSv2, and checkov/tflint/gitleaks scans. Documented 18 architecture decisions and added Terraform tests with mock providers.

Stack: AWS, Terraform, Karpenter, Istio, ArgoCD, Helm, Prometheus, Grafana, k6, Go, React.

Lemma — DevOps AI Agents Control Plane

- Built an AI-native operations platform for provisioning AWS infrastructure and deploying applications through natural-language agents, with FastAPI/SQLAlchemy and Next.js/TypeScript on PostgreSQL and Redis.
- Implemented queue-driven AWS sub-account bootstrap for isolated workspaces, with moto-mocked unit tests and gated AWS integration tests.
- Delivered a ZIP-to-live-URL pipeline: FastAPI on Fargate behind a per-workspace shared ALB, images built by CodeBuild in the customer’s account, and lazy provisioning to keep idle accounts near \$0/month.
- Engineered a Claude-powered deploy agent calling deterministic, tested provision/deploy/teardown tools, with human approval before consequential actions.

Stack: Python, FastAPI, SQLAlchemy, TypeScript, Next.js, AWS Fargate/ECS, ALB, CodeBuild, PostgreSQL, Redis, Docker, GitHub Actions, Claude.

TECHNICAL SKILLS

Cloud	AWS: ECS, EKS, EC2, Lambda, RDS, DocumentDB, DynamoDB, S3, CloudFront, VPC, Route 53, MSK, SQS; Microsoft Azure
Containers	Docker, Docker Swarm, Kubernetes, EKS, Helm, Portainer
CI/CD & IaC	Azure Pipelines, Azure Repos, GitHub Actions, Jenkins, GitLab, ArgoCD, GitOps, Terraform
Observability	Grafana, Loki, Prometheus, New Relic, CloudWatch
Data & security	PostgreSQL, MySQL, RDS, Redis, SonarQube, Trivy, IAM, DevSecOps
Systems	Linux (Ubuntu), WSL2, Bash

EDUCATION

Gogte Institute of Technology, India

Aug 2019 – May 2023

Bachelor of Engineering in Computer Science and Engineering · CGPA: 9.51/10